

Determining the Crystalline Structure of Metals through X-Ray Diffraction

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Abstract

This experiment set out to measure X-ray diffraction patterns of various solid samples in order to describe their crystal structure at the atomic/molecular level. An already calibrated Siemen !!!!!MODEL NAME!!!! X-Ray diffractometer was used to send X-rays into Copper (Cu), Nickel (Ni), CuNi alloy, Titanium (Sn), Lead (Pb), PbSn alloy and two mystery samples named A and B.

Any crystalline solid consists of a regular 3-dimensional packing of cells, the shape of which corresponds to one of fourteen Bravais lattices. These 3D patterns create multiple parallel planar layers within the crystal of various orientations. X-Rays travelling through these planes will diffract off the atoms and constructively interfere at the detector, creating a peak in photon count in the observed diffraction pattern. From that peak, Bragg's law $n\lambda = 2d \sin \theta$ (where $n = 1$ for a first-order plane) can be used to deduce the inter-planar distance d of the diffraction plane of the crystal. Using this inter-planar distance and the corresponding Miller indices, the Bravais lattice can be determined and the lattice parameter(s) calculated. The Cu and Ni pure samples and alloys were found to have a Face-Centered Cubic (FCC) Bravais lattice, with single lattice parameters [...] respectively. The alloy samples of these two was found to have [...]. The Sn and Pb samples were found to have a lattice of [...] and parameters [...] respectively, and their alloy [...], [...]. Finally, the mystery samples A and B were determined to be [...] and [...], with lattices [...] and parameters [...], respectively.

Contents

1	Introduction	1
2	Methods	1
3	Results	2
4	Discussion	3
5	Conclusion	4

6	Appendix	4
6.1	Inter-planar distance, Bravais Lattices and Miller indices	4
6.2	Supplementary Figures and Tables	5

1 Introduction

A solid can serve as a diffraction grating for X-rays of wavelengths similar to the lengths of inter-atomic distances. Similarly to how light diffracts, X-rays can also diffract off of atoms. Atoms within solids will arrange themselves in symmetries, formally called cells, given by one of one of fourteen Bravais lattices, with each lattice grouped into one of seven crystal systems. Atoms will also align along different diffraction planes depending on their Bravais lattice. If an X-ray is sent to a solid with an angle θ such that constructive interference occurs at a detector placed 2θ with respect to the incident ray, a peak in intensity of photons may be observed. From this, one can obtain a relation between the inter-planar distance d , the incident angle θ , and the wavelength λ :

$$n\lambda = 2d\sin(\theta) \tag{1}$$

with $n = 1$ for our purposes. Equation 1 is formally called Bragg's Law.

Using these observed peaks and (1), the distance d between the parallel diffraction planes can be obtained and related to the Bravais lattice and Miller indices by various equations. For a cubic system, this relation takes the form of

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2} \tag{2}$$

where h, k, l are the Miller Indices and a is the length of the cube edge. Similar equations for different crystal systems are given in 6.1.

2 Methods

The apparatus consisted of a pre-calibrated Siemens diffractometer, which consisted of an X-ray tube, a sample mounting plate, and a photon detector. The X-ray tube used Copper K_α radiation. Two sets of filters of 0.6 mm were present on the X-ray tube and the detector. During data acquisition, the diffractometer was set to 40 kV and 40 mA. The mounting plate kept the sample in place with a spring-loaded mechanism, and was marked with a red line

to center the samples. A computer connected to the apparatus was used to give it a .dql measurement parameter file and obtain the data as a .RAW file (subsequently converted to .UXD.) The measuring parameters were an initial angle of $2\theta_0 = 5.00^\circ$, angle increments of $\Delta\theta = 0.05^\circ$, with 2.00s at each step, with a final angle of $2\theta_f = 110.00^\circ$. Data acquisition consisted in mounting the sample on the plate, opening the parameter file in the computer software and executing the job.

3 Results

Three trials were done for Ni and Cu samples, though the data for the first Cu trial did not save correctly. For each of the alloys (including different percentage of mixing) one trial was done.

The fitted gaussian sums with residuals for the Cu and Ni samples and alloys are shown.

In figure 2, the residuals for the Ni samples are shown, and from their numerical results in table ?? we can observe that random error from trial to trial is minimal compared with noise error provided by the Gaussian fits. The fitted gaussian sums with residuals for the Cu and Ni samples and alloys are shown in the same appendix, in figure 3. From these, the inter-planar distances were calculated and are shown in table 1. These were plotted against expected results for all 3 types of cubic Bravais lattices, and the best fit was found to be the Face-Centered Cubic (FCC) lattice, as shown below in figure 1. The lattice parameters were then calculated and are shown in table 1.

Sample	a_{exp} (Å)	a_{theory} (Å)	Δa (Å)
Ni	3.5223 ± 0.0001	3.5238	-0.0015 ± 0.0001
Cu	3.6126 ± 0.0001	3.6149	-0.0023 ± 0.0001
Cu ₂₅ Ni ₇₅	3.5415 ± 0.0002	3.5467	-0.0052 ± 0.0002
Cu ₅₀ Ni ₅₀	3.5629 ± 0.0003	3.5696	-0.0067 ± 0.0003
Cu ₇₅ Ni ₂₅	3.5864 ± 0.0002	3.5924	-0.0060 ± 0.0002

Table 1: Experimental and theoretical lattice parameters for Cu, Ni pure samples and alloys. The theoretical values for the alloys are calculated using Vegard's law .

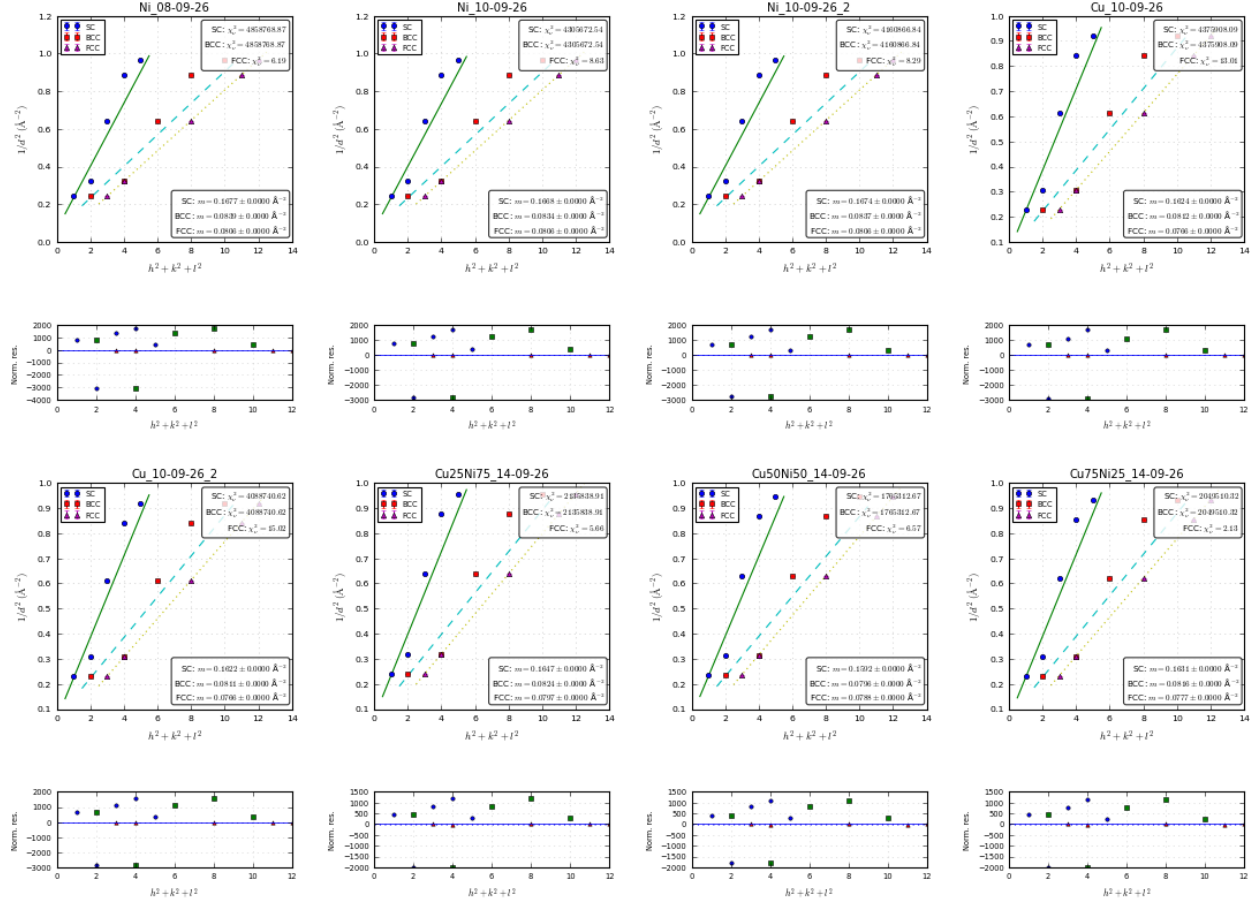


Figure 1: Determination of Cubic crystal structure and lattice parameter for Cu and Ni.

4 Discussion

The results for the Cu and Ni pure samples and alloys are consistent with the literature values, with the experimental values being within 0.2% of the theoretical values. The small discrepancies can be attributed to background noise in the acquired data. The determination of the Face-Centered Cubic (FCC) Bravais lattice for these samples is also consistent with known crystal structures for Cu and Ni. While running multiple trials does not reduce error much, it could still be worth it, as random error is of similar order as the noise error. The results for the Pb, Sn, and PbSn alloys are still to be determined, as well as the mystery samples A and B.

5 Conclusion

6 Appendix

6.1 Inter-planar distance, Bravais Lattices and Miller indices

For the following equations: d is the inter planar distance, a, b, c the inter-atomic distances, h, k, l the indices for a given Miller index (hkl).

Cubic:

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Tetragonal:

$$\frac{1}{d^2} = \frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}$$

Hexagonal:

$$\frac{1}{d^2} = \frac{4}{3} \left(\frac{h^2 + k^2 + hk}{a^2} \right) + \frac{l^2}{c^2}$$

Orthorhombic:

$$\frac{1}{d^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

Vegard's law:

$$a_{\text{alloy}} = xa_{Cu} + (1 - x)a_{Ni}$$

6.2 Supplementary Figures and Tables

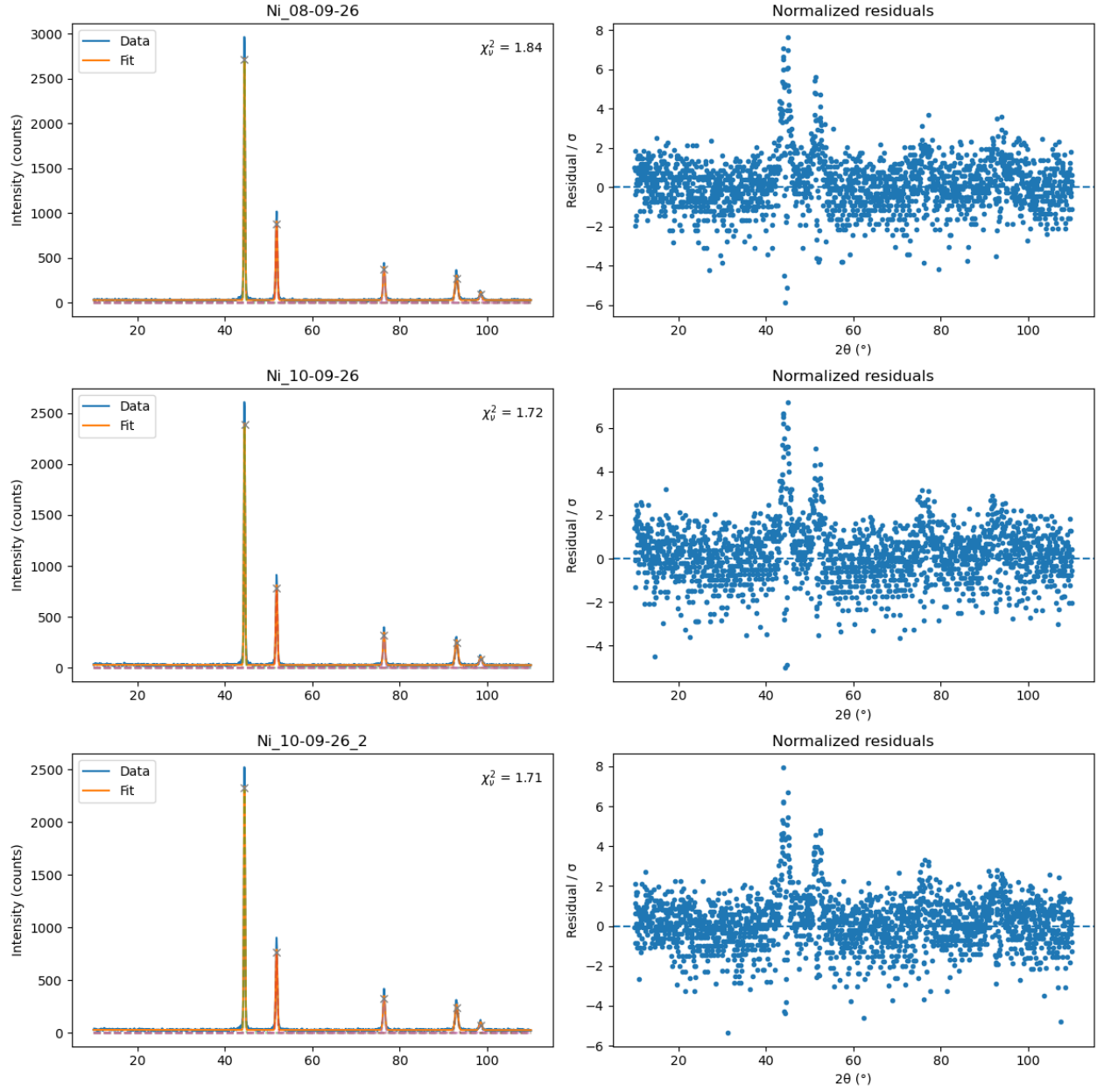


Figure 2: Gaussian sum fits of all pure Ni trials with residuals

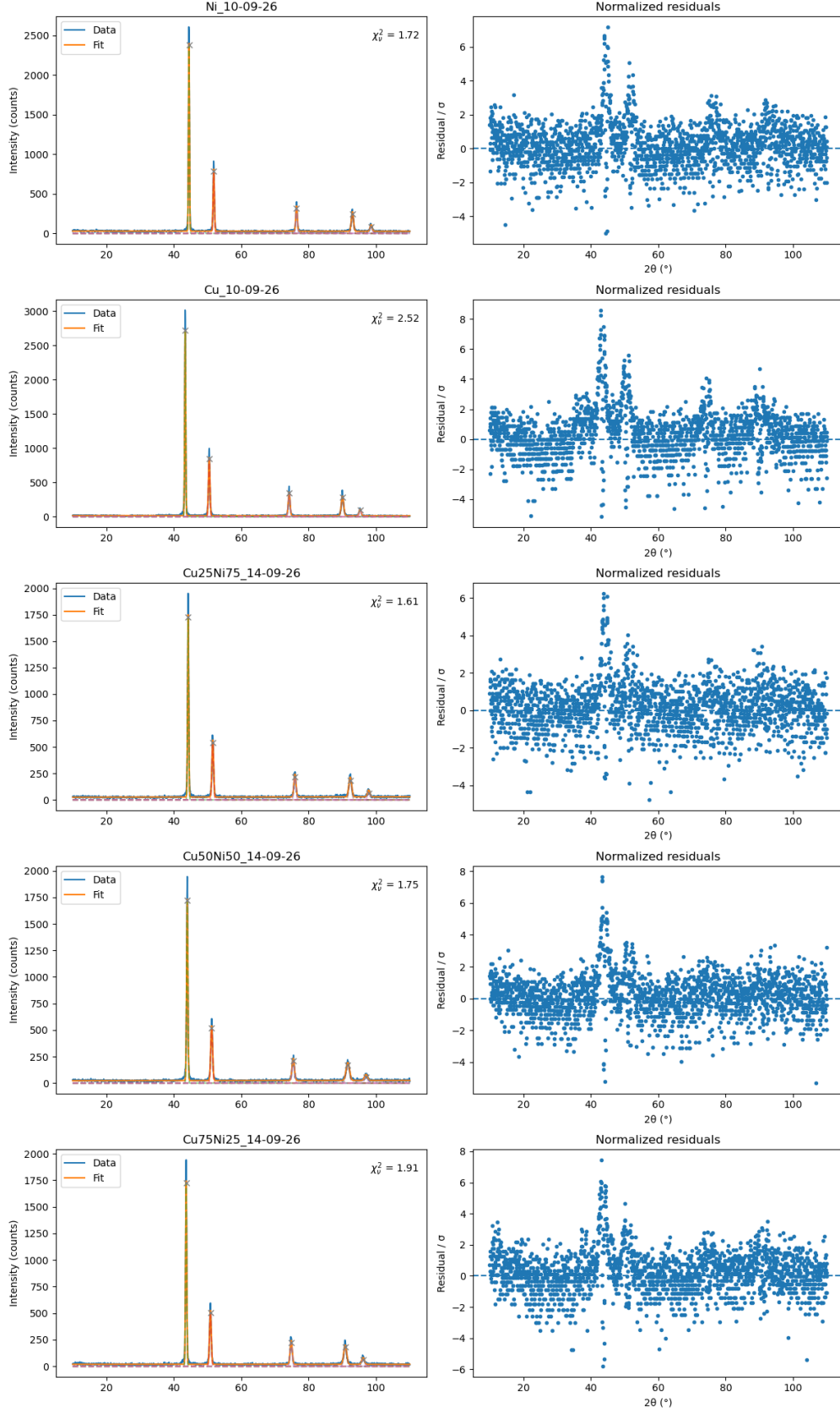


Figure 3: Gaussian sum fits with residuals of the Cu and Ni pure and alloy samples

Ni trial 1			Ni trial 2		
Peak	2θ (°)	d (Å)	Peak	2θ (°)	d (Å)
1	44.4819 ± 0.0011	2.03512 ± 0.00005	1	44.4843 ± 0.0011	2.03502 ± 0.00005
2	51.8330 ± 0.0023	1.76246 ± 0.00007	2	51.8331 ± 0.0024	1.76245 ± 0.00008
3	76.3993 ± 0.0041	1.24563 ± 0.00006	3	76.3933 ± 0.0045	1.24571 ± 0.00006
4	92.9684 ± 0.0059	1.06221 ± 0.00005	4	92.9701 ± 0.0062	1.06220 ± 0.00005
5	98.4832 ± 0.0109	1.01694 ± 0.00008	5	98.5025 ± 0.0127	1.01679 ± 0.00010
Ni trial 3			Cu trial 1		
Peak	2θ (°)	d (Å)	Peak	2θ (°)	d (Å)
1	44.4810 ± 0.0012	2.03517 ± 0.00005	1	43.3661 ± 0.0011	2.08487 ± 0.00005
2	51.8307 ± 0.0024	1.76253 ± 0.00008	2	50.4888 ± 0.0024	1.80618 ± 0.00008
3	76.4023 ± 0.0045	1.24558 ± 0.00006	3	74.2095 ± 0.0043	1.27687 ± 0.00006
4	92.9767 ± 0.0062	1.06214 ± 0.00005	4	90.0228 ± 0.0054	1.08915 ± 0.00005
5	98.4665 ± 0.0135	1.01707 ± 0.00010	5	95.2553 ± 0.0097	1.04266 ± 0.00008
Cu trial 2			Cu ₂₅ Ni ₇₅		
Peak	2θ (°)	d (Å)	Peak	2θ (°)	d (Å)
1	43.3667 ± 0.0011	2.08484 ± 0.00005	1	44.2433 ± 0.0015	2.04554 ± 0.00006
2	50.4882 ± 0.0024	1.80620 ± 0.00008	2	51.5406 ± 0.0034	1.77176 ± 0.00011
3	74.2000 ± 0.0043	1.27701 ± 0.00006	3	75.9226 ± 0.0065	1.25226 ± 0.00009
4	90.0104 ± 0.0057	1.08927 ± 0.00005	4	92.3239 ± 0.0085	1.06793 ± 0.00008
5	95.2715 ± 0.0098	1.04253 ± 0.00008	5	97.7855 ± 0.0166	1.02232 ± 0.00013
Cu ₅₀ Ni ₅₀			Cu ₇₅ Ni ₂₅		
Peak	2θ (°)	d (Å)	Peak	2θ (°)	d (Å)
1	43.9763 ± 0.0016	2.05734 ± 0.00007	1	43.6499 ± 0.0015	2.07197 ± 0.00007
2	51.2219 ± 0.0036	1.78204 ± 0.00012	2	50.8455 ± 0.0035	1.79434 ± 0.00011
3	75.4112 ± 0.0071	1.25948 ± 0.00010	3	74.8037 ± 0.0062	1.26819 ± 0.00009
4	91.6044 ± 0.0098	1.07443 ± 0.00009	4	90.8289 ± 0.0083	1.08157 ± 0.00008
5	97.0016 ± 0.0200	1.02849 ± 0.00016	5	96.1327 ± 0.0160	1.03546 ± 0.00013

Table 2: Measured X-ray diffraction peak angular positions and corresponding interplanar spacings (d) for the Ni, Cu, and Cu–Ni alloy samples.